

# VTVL Hopper Engineering Notebook

## Requirements and Propellant Selection

### 1 Design Requirements

#### 1.1 Mission Objective

Design a throttleable, pressure-fed liquid bipropellant engine for a minimal VTVL hopper. The vehicle exists to support the engine, tanks, avionics, plumbing, landing gear, and test instrumentation. There is no payload requirement and no altitude requirement at this stage. The first flight objective is controlled liftoff, hover or near-hover control authority, and safe landing.

#### 1.2 Requirement Classification

The requirements below are limited to items that must be true before the design is considered acceptable. Quantities that depend on vehicle mass convergence, tank selection, injector pressure drop, cooling margin, or throttle hardware are carried as sizing variables, not fixed requirements.

Table 1: Design requirements for the first prototype phase.

| Item                   | Classification     | Requirement / design rule                                                                                                                                                                             |
|------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Budget                 | Hard               | Prototype propulsion hardware target: $\leq \$5,000$ excluding propellant, shop tooling, and instrumentation already owned.                                                                           |
| Architecture           | Hard               | Pressure-fed liquid bipropellant. No turbopumps.                                                                                                                                                      |
| Cryogenics             | Hard               | No cryogenic propellant storage for the first prototype.                                                                                                                                              |
| Hazard class           | Hard               | No hydrazine-family fuels, NTO, RFNA, or similarly high-toxicity/high-corrosion propellants for this build.                                                                                           |
| Material access        | Hard               | Primary wetted structural materials shall be readily obtainable aluminum alloy or stainless steel unless a later section gives a specific compatibility basis.                                        |
| Fluid connectors       | Hard               | Use pressure-rated commercial connector families only: AN/MS, CGA, Swagelok-compatible compression fittings, or equivalent documented hardware. No custom fluid connector geometry.                   |
| Pressure safety        | Hard               | Pressure-wetted custom parts shall use burst factor of safety $\geq 4$ against MEOP. Purchased cylinders, valves, regulators, and fittings shall be used only within documented manufacturer ratings. |
| Remote operation       | Hard               | All oxidizer and hot-fire operations shall be controllable from a safe remote location with positive propellant isolation before personnel approach.                                                  |
| Throttle demonstration | Test requirement   | The engine shall demonstrate repeatable operation at full thrust and at lower commanded setpoints before flight. The final throttle ratio is set after vehicle mass is known.                         |
| Combustion stability   | Test requirement   | No sustained hard-start, chugging, buzz, or high-frequency instability is acceptable in qualification hot-fire data. Stability criteria are finalized in the injector and test sections.              |
| TVC interface          | Layout requirement | Reserve a defined thrust-frame or gimbal interface. Cooling jacket and plumbing geometry shall not block later TVC integration.                                                                       |

Table 2: Sizing variables intentionally deferred to later notebook sections.

| Quantity                   | Status          | How it is used in this notebook                                                                                                                                                                |
|----------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Specific impulse           | Analysis output | Report $I_{sp,SL}$ , $I_{sp,vac}$ , $P_c$ , $P_{amb}$ , $P_e$ , $\epsilon$ , O/F, and chemistry mode for every CEA/RocketCEA value. No hard $I_{sp}$ floor is set in the requirements section. |
| Nominal thrust             | Sizing variable | Preliminary analyses may bracket 200-400 N, but the final value belongs in the engine sizing section after dry mass and propellant mass converge.                                              |
| Full-thrust liftoff margin | Derived later   | After $m_0$ is known, require $\frac{T_{max}}{m_0 g_0} \geq 1.2$ minimum. Higher margin is useful only if it does not force unnecessary system scale-up.                                       |
| Minimum throttle           | Derived later   | After $m_0$ is known, require $T_{min} < m_0 g_0$ so the vehicle can command descent.                                                                                                          |
| Burn duration              | Design goal     | 30 s minimum useful demonstration; 60 s stretch target. Propellant mass is computed after thrust and $I_{sp,SL}$ are selected.                                                                 |
| Chamber pressure           | Design variable | Use 150 psia as the first static-fire analysis point and carry 200 psia as a performance-upgrade case if feed, cooling, and structure close.                                                   |
| Engine dry mass            | Design goal     | Engine assembly target $\leq 2.5$ kg for chamber, nozzle, injector, and cooling jacket.                                                                                                        |

### 1.3 Chamber Pressure Design Basis

The baseline chamber-pressure range is  $P_c = 150 - 200$  psia. A lower chamber pressure reduces chamber stress, heat flux, valve pressure rating, injector pressure drop, and early test-stand load. A higher chamber pressure improves nozzle pressure ratio and sea-level impulse. For this hopper, the first static-fire baseline is  $P_c = 150$  psia;  $P_c = 200$  psia is retained as a later upgrade point, not assumed from the start.

The generated RocketCEA pressure sweep in rocket\_outputs/data/pressure\_sweep\_summary.csv gives the following sea-level-optimized performance trend [1]. Each row re-optimizes O/F and  $\epsilon$  so that  $P_e \approx P_{amb}$  at  $P_{amb} = 14.7$  psia.

Table 3: Sea-level pressure sensitivity for finalist propellants. Values are generated by the included RocketCEA script and should be regenerated if the propellant model, ambient pressure, or chemistry mode changes.

| Propellant                 | $I_{sp,SL}$ at 150 psia | $I_{sp,SL}$ at 200 psia | Gain   |
|----------------------------|-------------------------|-------------------------|--------|
| N <sub>2</sub> O / Ethanol | 204.8s                  | 215.5s                  | +10.6s |
| N <sub>2</sub> O / IPA     | 207.0s                  | 217.7s                  | +10.7s |
| GOX / Ethanol              | 225.6s                  | 237.8s                  | +12.2s |
| GOX / IPA                  | 230.1s                  | 242.4s                  | +12.4s |

## 2 Propellant

### 2.1 Problem

Select a practical non-cryogenic oxidizer/fuel pair for a low-altitude throttleable hopper. The selected pair should minimize development risk and stored-energy complexity while still keeping propellant mass, tank size, injector design, cooling, ignition, and operating procedures tractable.

### 2.2 Thermochemical Method and Evidence Standard

All thermochemical values in this section use the included script optimize\_cea.py and the generated output package [1]. The script wraps RocketCEA, which is a Python interface to NASA CEA. NASA CEA computes chemical-equilibrium rocket performance and supports equilibrium and frozen expansion modes [2]. RocketCEA documents get\_Isp as vacuum impulse and estimate\_Ambient\_Isp as the function that applies a specified ambient pressure and reports the operating mode [3], [4].

The design-basis comparison uses:

- $P_c = 150$  psia.
- $P_{\text{amb}} = 14.7$  psia.
- Shifting-equilibrium chemistry.
- O/F swept over a broad candidate-specific range.
- At each O/F,  $\varepsilon$  chosen so that  $P_e \approx P_{\text{amb}}$ .
- Sea-level impulse computed with `estimate_Ambient_Isp`, not `vacuum_get_Isp`.

#### ▷ Performance-value rule

A CEA/RocketCEA performance number is incomplete unless it states  $P_c$ ,  $P_{\text{amb}}$ ,  $\varepsilon$ ,  $P_e$ , O/F, chemistry mode, and whether the reported impulse is  $I_{\text{sp,SL}}$  or  $I_{\text{sp,vac}}$ . In this notebook, propellant selection uses sea-level ambient-corrected  $I_{\text{sp,SL}}$ .

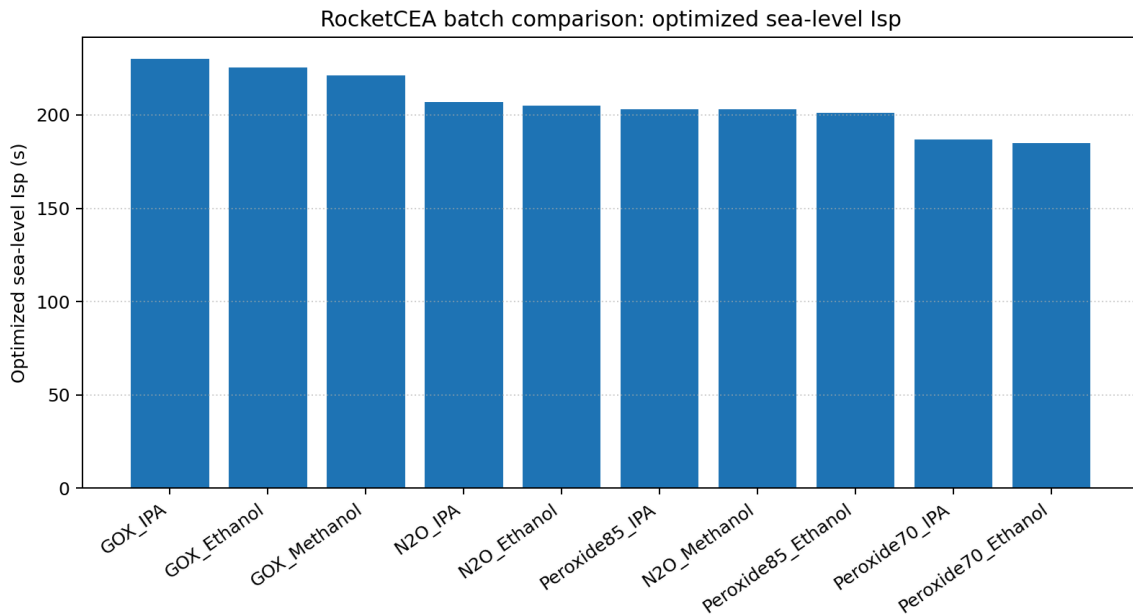


Figure 1: Generated RocketCEA comparison for optimized sea-level impulse at  $P_c = 150$  psia,  $P_{\text{amb}} = 14.7$  psia, shifting-equilibrium chemistry, and  $P_e \approx P_{\text{amb}}$  nozzle sizing [1].

## 2.3 Candidate Survey

### 2.3.1 Hard-Eliminated Candidates

These combinations are removed before finalist comparison because a safety, legality, sourcing, or architecture constraint is sufficient by itself.

- **LOX-based propellants** (LOX/ethanol, LOX/RP-1, LOX/LCH<sub>4</sub>, LOX/LH<sub>2</sub>): eliminated because the first prototype forbids cryogenic storage.
- **Hydrazine-family fuels, MMH, UDMH, and NTO**: eliminated for toxicity, regulatory burden, and handling risk.
- **RFNA / ethanol**: eliminated for toxicity, corrosion, and incompatibility with the low-cost aluminum/stainless construction path.
- **High-test peroxide at 90%+**: eliminated for the first prototype because the project does not yet have the sourcing, cleaning, catalyst-bed, compatibility, and decomposition-risk controls required for this oxidizer class [5].

- **N<sub>2</sub>O / RP-1 or kerosene:** eliminated for this regen-cooled first prototype because fuel coking and residue management are unnecessary risks when alcohol fuels are available.

### 2.3.2 Performance-Eliminated or Deprioritized Candidates

Table 4: Sea-level performance screening for non-finalist liquid candidates. Conditions:  $P_c = 150$  psia,  $P_{amb} = 14.7$  psia, shifting-equilibrium chemistry,  $P_e \approx P_{amb}$  [1].

| Candidate                                   | O/F  | $\epsilon$ | $I_{sp,SL}$ | Disposition                                                                                                               |
|---------------------------------------------|------|------------|-------------|---------------------------------------------------------------------------------------------------------------------------|
| 70% H <sub>2</sub> O <sub>2</sub> / Ethanol | 6.30 | 2.30       | 185.0s      | Eliminated: low sea-level performance plus peroxide handling complexity.                                                  |
| 70% H <sub>2</sub> O <sub>2</sub> / IPA     | 7.20 | 2.31       | 186.9s      | Eliminated: low sea-level performance plus peroxide handling complexity.                                                  |
| 85% H <sub>2</sub> O <sub>2</sub> / Ethanol | 5.00 | 2.36       | 201.1s      | Deprioritized: similar to N <sub>2</sub> O/alcohol impulse but with peroxide-specific catalyst and decomposition hazards. |
| 85% H <sub>2</sub> O <sub>2</sub> / IPA     | 5.65 | 2.37       | 203.2s      | Deprioritized: no clear advantage over N <sub>2</sub> O/alcohol for a first hopper.                                       |
| N <sub>2</sub> O / Methanol                 | 3.50 | 2.38       | 203.0s      | Deprioritized: below N <sub>2</sub> O/IPA and adds methanol toxicity.                                                     |
| GOX / Methanol                              | 1.20 | 2.46       | 221.3s      | Deprioritized: below GOX/IPA and GOX/ethanol with no practical offsetting advantage.                                      |

**H<sub>2</sub>O<sub>2</sub> at 80-85% / ethanol or IPA.** This concentration is more accessible than 90%+ peroxide, but it still requires peroxide-compatible storage, cleaning, decomposition controls, and usually catalyst hardware. The 265 s value sometimes quoted from peroxide monopropellant development literature is a high-expansion-ratio vacuum monopropellant number and does not apply to this low-expansion sea-level bipropellant comparison [5]. At the design-basis condition, 85% peroxide/alcohol does not materially outperform N<sub>2</sub>O/alcohol and is not the lowest-risk path.

**Methanol combinations.** Methanol has legitimate historical rocket use, including as part of German C-Stoff mixtures, but it does not outperform ethanol or IPA in the design-basis RocketCEA comparison. It also has a worse handling profile; OSHA lists a 200 ppm permissible exposure limit for methyl alcohol [6]. Methanol remains technically possible, but it does not merit a finalist slot.

**N<sub>2</sub>O with gaseous hydrocarbon fuels.** Ethylene, ethane, and propane can be made to work in propulsion systems, but they are a poor match to this ground hopper. Ethylene has a critical temperature of about 282.5K (9.3°C), so it cannot be stored as a normal room-temperature liquid; NIST lists ethylene critical data near  $T_c = 282.5$ K and  $P_c = 50.6$  bar [7]. Propane is easier to store, but its room-temperature vapor pressure is not enough to feed a  $P_c = 150$  psia engine without additional pressurization [8]. Liquid alcohol fuels win on density, sourcing, cooling usefulness, and injector simplicity.

## 2.4 The Four Viable Finalists

The four viable first-prototype finalists are N<sub>2</sub>O/IPA, N<sub>2</sub>O/ethanol, GOX/IPA, and GOX/ethanol. They are not equivalent: GOX/alcohol has better design-basis sea-level performance and easier single-phase injector modeling, while N<sub>2</sub>O/alcohol has compact self-pressurizing oxidizer storage and direct small-hopper precedent.

Table 5: Design-basis RocketCEA performance for finalist propellants. Conditions:  $P_c = 150$  psia,  $P_{amb} = 14.7$  psia, shifting-equilibrium chemistry,  $P_e \approx P_{amb}$  [1].

| Propellant                 | O/F  | $\epsilon$ | $I_{sp,SL}$ | $I_{sp,vac}$ | $T_c$ | $c^*$    |
|----------------------------|------|------------|-------------|--------------|-------|----------|
| GOX / IPA                  | 1.65 | 2.43       | 230.1s      | 272.9s       | 3237K | 1762 m/s |
| GOX / Ethanol              | 1.55 | 2.45       | 225.6s      | 268.0s       | 3187K | 1727 m/s |
| N <sub>2</sub> O / IPA     | 5.10 | 2.37       | 207.0s      | 244.6s       | 3120K | 1590 m/s |
| N <sub>2</sub> O / Ethanol | 4.65 | 2.38       | 204.8s      | 242.2s       | 3064K | 1573 m/s |

### Key observation

At  $P_c = 150$  psia and sea-level ambient pressure, the GOX/alcohol options are about 19-25 s ahead of the  $N_2O$ /alcohol options in ideal ambient-corrected impulse. That advantage is real, but it must be traded against compressed-oxygen storage mass, regulator cost, oxygen cleaning, and fire safety. For a barely-flying hopper, feed-system risk can dominate ideal  $I_{sp}$ .

#### 2.4.1 Common $N_2O$ Architecture

$N_2O$  is stored as a saturated liquid at its own vapor pressure. Resonac lists nitrous oxide vapor pressure at 20°C as 5.24 MPa and gives a critical temperature of 36.41°C with critical pressure 7.24 MPa [9]. This allows compact self-pressurizing oxidizer storage without a separate high-pressure inert pressurant.

The disadvantage is that feed pressure is temperature-dependent. The oxidizer tank, injector pressure drop, oxidizer mass flow, and thrust map change with tank temperature and with evaporative cooling during discharge. Any  $N_2O$  engine must therefore treat tank temperature and pressure as control inputs, not background conditions.

##### 2.4.1.1 Injector: Two-Phase Flow

$N_2O$  can flash through restrictions because it is often near saturation before the injector. The standard incompressible single-phase orifice equation,

$$\dot{m} = C_d A \sqrt{2\rho\Delta P}$$

is useful only as an upper-bound starting point for the oxidizer injector. Dyer-style non-homogeneous non-equilibrium modeling, homogeneous equilibrium modeling, and cold-flow/hot-fire calibration are needed for credible final sizing [10], [11], [12], [13].

Table 6:  $N_2O$  injector mass-flow model hierarchy.

| Model       | Assumption                                                   | Expected bias                                                                        | Use in this project                                           |
|-------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------|
| SPI         | Single-phase incompressible liquid.                          | Often overpredicts $N_2O$ oxidizer flow after flashing starts.                       | Upper-bound geometry estimate only.                           |
| HEM         | Two-phase thermodynamic equilibrium through the restriction. | Can underpredict short-orifice flow where equilibrium is not reached.                | Lower-bound check and long-channel reference.                 |
| Dyer / NHNE | Non-equilibrium transition between SPI and HEM.              | Most practical published engineering model for self-pressurizing oxidizer injectors. | Primary $N_2O$ injector sizing model before test calibration. |

The practical  $N_2O$  injector workflow is:

1. Size an initial oxidizer orifice using SPI.
2. Compute HEM for the same geometry to establish a lower-bound flow estimate.
3. Apply Dyer/NHNE or a comparable two-phase model using the selected orifice  $\frac{L}{D}$ .
4. Cold-flow the injector over the expected tank-temperature range.
5. Update the throttle map using hot-fire data.

##### 2.4.1.2 Throttling Behavior

The AEL  $N_2O$ /IPA throttle work is directly relevant because it studied a small closed-loop throttleable VTVL thruster at approximately the right architecture. The reported behavior was dominated by  $N_2O$  phase change through the control valve and injector rather than by simple hydraulic pressure-drop scaling [14]. This makes empirical throttle-map calibration mandatory for a flight hopper.

### 2.4.1.3 Material Compatibility and Operations

N<sub>2</sub>O hardware must be kept oxygen-clean enough to avoid fuel contamination on the oxidizer side. Aluminum and stainless hardware can be used when the system is designed around compatible seals, cleaned assembly practice, relief devices, and conservative temperature control. The Las Vegas temperature case is handled separately in Section 2.6.

### 2.4.2 N<sub>2</sub>O / IPA

**Verdict: Finalist** N<sub>2</sub>O/IPA is the leading first-hopper selection. It is not the highest- $I_{sp}$  option, but it combines compact non-cryogenic oxidizer storage, easy fuel sourcing, direct VTVL precedent, and acceptable sea-level performance.

Design-basis RocketCEA result:  $I_{sp,SL} = 207.0s$ ,  $I_{sp,vac} = 244.6s$ , O/F = 5.10,  $\varepsilon = 2.37$ ,  $T_c = 3120K$ , and  $c^* = 1590$  m/s.

**Flight heritage.** GSP's Colibri vehicle used a bipropellant engine with N<sub>2</sub>O and IPA. Kistler reports that Colibri is a 2.45m reusable VTVL demonstrator with up to 1.25 kN thrust; by October 2024 it had completed 53 flights with safe landings [15]. European Spaceflight reports a 105 m free flight on 18 October 2024, a 30 m lateral translation, and a 60 s flight time [16]. This is the closest documented analog to the intended project.

**Fuel sourcing.** 99% IPA is widely available through hardware, janitorial, and chemical suppliers. 70% disinfectant is not suitable because the water content changes performance and ignition behavior.

**Cooling.** IPA has slightly higher room-temperature specific heat than ethanol, while ethanol has higher thermal conductivity. CRC values used here are  $c_p \approx 2.68$  kJ/kg/K and  $k \approx 0.135$  W/m/K for IPA at room temperature [17]. Both alcohols remain plausible regenerative coolants at this scale; final margin belongs in the cooling section.

### 2.4.3 N<sub>2</sub>O / Ethanol

**Verdict: Finalist** N<sub>2</sub>O/ethanol is the strongest N<sub>2</sub>O backup path because it has a larger academic static-test literature base and slightly better thermal conductivity as a coolant.

Design-basis RocketCEA result:  $I_{sp,SL} = 204.8s$ ,  $I_{sp,vac} = 242.2s$ , O/F = 4.65,  $\varepsilon = 2.38$ ,  $T_c = 3064K$ , and  $c^* = 1573$  m/s.

**Experimental literature.** NMIMT/Sandia conducted N<sub>2</sub>O/ethanol engine development and test work, and ISAS/JAXA studied a 2 kN-class N<sub>2</sub>O/ethanol propulsion system [18], [19], [20]. These sources support the credibility of N<sub>2</sub>O/ethanol combustion development, but their reported impulse values must not be transferred blindly unless  $P_c$ ,  $P_{amb}$ , nozzle expansion, and data-reduction basis match this design.

**Cooling.** Ethanol's room-temperature thermal conductivity is about 0.171 W/m/K versus about 0.135 W/m/K for IPA, while its specific heat is slightly lower, about 2.57 kJ/kg/K versus 2.68 kJ/kg/K [17]. This makes ethanol attractive if the cooling jacket is heat-flux limited rather than bulk-temperature-rise limited.

**Sourcing.** Anhydrous ethanol requires more procurement control than IPA. 190-proof ethanol contains enough water to change performance and optimum O/F, so 200-proof or otherwise documented water content is required for repeatable CEA-to-test comparison.

### 2.4.4 Common GOX Architecture

GOX provides the cleanest injector physics: the oxidizer can be treated as a regulated single-phase gas upstream of the injector, so the first-pass orifice model is much more predictable than N<sub>2</sub>O flashing flow. The trade is tankage and operations. A useful GOX system requires a high-pressure oxygen cylinder or lightweight vessel, a high-flow oxygen-compatible regulator, oxygen-clean assembly practice, and strict separation between oxygen-side and fuel-side tooling.

For equal impulse, GOX generally needs substantially more oxidizer storage volume than liquid  $\text{N}_2\text{O}$ . The ideal-gas oxygen volume estimate is:

$$V_{\text{GOX}} = \frac{m_{\text{O}_2} R_{\text{O}_2} T}{P}.$$

At 2200 psia and 300 K, the gas volume for a few kilograms of oxygen is already tens of liters. The equivalent  $\text{N}_2\text{O}$  mass stores as a dense saturated liquid, typically giving a much smaller tank volume. This storage-volume penalty is the main reason GOX is not automatically selected despite its better sea-level impulse.

### 2.4.5 GOX / IPA

**Verdict: Finalist** GOX/IPA is the design-basis performance leader among the non-cryogenic candidates analyzed here, but it carries the compressed-oxygen storage and regulator burden.

Design-basis RocketCEA result:  $I_{\text{sp,SL}} = 230.1\text{s}$ ,  $I_{\text{sp,vac}} = 272.9\text{s}$ ,  $\text{O/F} = 1.65$ ,  $\varepsilon = 2.43$ ,  $T_c = 3237\text{K}$ , and  $c^* = 1762\text{ m/s}$ .

GOX/IPA deserves a full design pass if the project can source oxygen-compatible regulators, vessels, valves, fittings, and cleaning procedures inside the budget. Its weakness is not performance; it is system mass, system cost, and oxygen safety practice.

### 2.4.6 GOX / Ethanol

**Verdict: Finalist** GOX/ethanol is nearly as strong as GOX/IPA analytically and has useful ignition literature at comparable chamber pressure.

Design-basis RocketCEA result:  $I_{\text{sp,SL}} = 225.6\text{s}$ ,  $I_{\text{sp,vac}} = 268.0\text{s}$ ,  $\text{O/F} = 1.55$ ,  $\varepsilon = 2.45$ ,  $T_c = 3187\text{K}$ , and  $c^* = 1727\text{ m/s}$ .

A 1984 NASA ignition characterization study tested GOX/ethanol near  $P_c = 150\text{ psia}$  and  $\text{O/F}$  near 1.8 [21]. That makes GOX/ethanol valuable as a lower-injector-risk fallback if  $\text{N}_2\text{O}$  two-phase design becomes the bottleneck.

## 2.5 Decision Analysis

### 2.5.1 Oxidizer-Level Trade: $\text{N}_2\text{O}$ vs. GOX

Table 7: Oxidizer-level architecture comparison.

| Criterion             | $\text{N}_2\text{O}$ path                                                     | GOX path                                                                   |
|-----------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Sea-level performance | Lower: about 205 – 207s at 150 psia.                                          | Higher: about 226 – 230s at 150 psia.                                      |
| Oxidizer storage      | Compact saturated liquid; self-pressurizing.                                  | Bulky compressed gas unless a custom lightweight vessel is used.           |
| Feed stability        | Tank pressure changes with temperature and discharge history.                 | Regulated feed pressure can be stable and repeatable.                      |
| Injector physics      | Two-phase flashing; requires Dyer/HEM/SPI bracketing and testing.             | Mostly single-phase gas; simpler first-pass analysis.                      |
| Hardware cost         | No oxygen regulator; pressure vessel still safety-critical.                   | High-flow oxygen regulator and oxygen-rated fittings are major cost items. |
| Operating risk        | Thermal management near $\text{N}_2\text{O}$ critical temperature is central. | Oxygen cleaning and combustion-contamination control are central.          |
| Best reason to choose | Compact, self-contained, directly relevant hopper precedent.                  | Higher $I_{\text{sp}}$ and easier injector sizing.                         |

GOX/IPA and GOX/ethanol are analytically attractive. They should win if the oxygen storage, regulator, cleaning, and mass model close. For the first low-complexity hopper path,  $\text{N}_2\text{O}$  remains more attractive because it avoids cryogenics and a regulated high-flow oxygen gas system while keeping the oxidizer tank compact.

### 2.5.2 Fuel-Level Trade Inside the $\text{N}_2\text{O}$ Path

Table 8:  $\text{N}_2\text{O}$ /IPA versus  $\text{N}_2\text{O}$ /ethanol.

| Criterion                       | $\text{N}_2\text{O}$ / IPA                   | $\text{N}_2\text{O}$ / Ethanol                           |
|---------------------------------|----------------------------------------------|----------------------------------------------------------|
| Design-basis $I_{\text{sp,SL}}$ | 207.0s                                       | 204.8s                                                   |
| Optimum O/F                     | 5.10                                         | 4.65                                                     |
| Flame temperature               | 3120K                                        | 3064K                                                    |
| Fuel access                     | Excellent: 99% IPA is widely available.      | Moderate: anhydrous ethanol sourcing must be controlled. |
| Coolant $c_p$                   | Slightly higher: 2.68 kJ/kg/K [17].          | Slightly lower: 2.57 kJ/kg/K [17].                       |
| Coolant thermal conductivity    | Lower: 0.135 W/m/K [17].                     | Higher: 0.171 W/m/K [17].                                |
| VTVL precedent                  | Direct: GSP Colibri and AEL work [14], [15]. | Strong static-test literature [18], [20].                |
| Primary advantage               | Sourcing and flight-relevant precedent.      | Academic literature and thermal conductivity.            |

#### 2.5.2.1 Pros and Cons

##### 2.5.2.1.1 $\text{N}_2\text{O}$ / IPA

###### Advantages

- Direct VTVL flight precedent at the closest public analog scale.
- IPA is inexpensive and easy to source at high purity.
- Slightly higher room-temperature  $c_p$  than ethanol.
- AEL throttle work is on  $\text{N}_2\text{O}$ /IPA.

###### Disadvantages

- Lower thermal conductivity than ethanol.
- Less academic combustion literature than  $\text{N}_2\text{O}$ /ethanol.
- All  $\text{N}_2\text{O}$  two-phase injector and temperature risks still apply.

##### 2.5.2.1.2 $\text{N}_2\text{O}$ / Ethanol

###### Advantages

- Stronger academic static-test literature base.
- Higher thermal conductivity as a regenerative coolant.
- Historically common alcohol rocket fuel.
- Good backup if following NMIMT/JAXA data closely.

###### Disadvantages

- Anhydrous ethanol procurement is less convenient.
- Water content must be controlled.
- No direct small VTVL flight precedent as strong as GSP Colibri.

### 2.5.3 Recommendation

**$\text{N}_2\text{O}$ /IPA is the primary first-hopper selection.** The reason is not maximum ideal impulse. The reason is total development risk: easy fuel sourcing, compact self-pressurizing oxidizer storage, and direct VTVL flight precedent outweigh a roughly 20-25 s ideal sea-level impulse advantage from GOX when the vehicle only needs to demonstrate controlled low-altitude flight.

**$\text{N}_2\text{O}$ /ethanol is the first backup.** Choose it if the cooling section shows ethanol's higher thermal conductivity is important, if anhydrous ethanol is already sourced, or if the design intentionally follows the NMIMT/JAXA  $\text{N}_2\text{O}$ /ethanol literature path.

**GOX/IPA or GOX/ethanol remain performance backups.** Choose GOX if the project prioritizes injector simplicity and can close the compressed-oxygen regulator, vessel, cleaning, and mass budget. If GOX is chosen, IPA has the slightly higher design-basis RocketCEA result while ethanol has the stronger ignition-literature tie through the NASA study.

## 2.6 Las Vegas Operational Considerations

### ▷ N<sub>2</sub>O critical temperature and MEOP definition

N<sub>2</sub>O has a critical temperature of 36.41°C and critical pressure of 7.24 MPa [9]. A N<sub>2</sub>O tank in a hot vehicle, direct sun, or summer test area can approach a very different pressure state than the same tank in a shaded room. Tank temperature is therefore part of the design basis, not just an operating note.

Design implications:

1. MEOP must be defined from the maximum credible filled-tank temperature.
2. Fill fraction, ullage, relief devices, and tank material selection must be documented before any hot-weather oxidizer loading.
3. Throttle maps must be temperature-indexed because N<sub>2</sub>O vapor pressure changes injector pressure drop and mass flow.
4. Test operations should standardize tank temperature or explicitly correct for it.

## 2.7 Propellant Section Next Actions

1. Commit batch\_optimum\_summary.csv and pressure\_sweep\_summary.csv with the notebook.
2. For the selected N<sub>2</sub>O/IPA baseline, run a dedicated CEA sweep after selecting actual fuel purity and expected chamber-pressure range.
3. Build the N<sub>2</sub>O injector model using SPI, HEM, and Dyer/NHNE bracketing before selecting orifice count and diameter.
4. Do not size final thrust or total propellant mass in this section; that belongs in the engine sizing and vehicle mass-convergence sections.

## 3 Engine Archetypes

## 4 Thermochemical Analysis

## 5 Engine Sizing

## 6 Nozzle Design

## 7 Injector Design

## 8 Feed System Design

## 9 Cooling Analysis

## 10 Structural Analysis

## 11 CAD Design

## 12 Simulation

## Bibliography

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